

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

GitHub Link :

Aim : Image Processing

Objective

To study and implement different gray-level transformation techniques for image enhancement, such as contrast stretching, histogram equalization, gamma transformation, log transformation, and image negative, and to analyze their effects using image and histogram comparisons.

Introduction

Gray-level transformation is an image enhancement technique used to improve the visual quality of an image.

In this technique, the intensity or brightness value of each pixel is changed to make the image clearer and to highlight important details.

For an 8-bit grayscale image, the intensity values range from 0 to 255.

- 0 represents black.
- 255 represents white.
- Values between 0 and 255 represent different shades of gray.

Contrast Stretching

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Contrast stretching is used to improve an image that has low contrast.

In a low-contrast image, the pixel values are usually concentrated in a small range. Because of this, the image may look dull or unclear.

Contrast stretching expands the range of pixel intensities.

Main effects

- **Dark areas become darker.**
- **Bright areas become brighter.**
- **Image contrast increases.**
- **Important details become easier to see.**
- **Overall image clarity improves.**

Therefore, contrast stretching is useful for making a low-contrast image more clear.

Histogram Equalization

A histogram shows how pixel intensity values are distributed in an image.

Histogram equalization is an image enhancement technique that improves contrast by redistributing the intensity values.

It is useful when some details are not clearly visible in the original image.

Main effects

- **Improves image contrast.**

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

- Redistributes pixel intensity values.
- Can reveal hidden details.
- Makes the image easier to analyze.

The original and enhanced histograms can be compared to understand how the intensity distribution has changed.

Histogram

A histogram is a graphical representation of the distribution of pixel intensity values in an image.

For a grayscale image:

- The horizontal axis represents pixel intensity values.
- The vertical axis represents the number of pixels having those intensity values.

For an 8-bit grayscale image, the intensity range is 0 to 255.

Importance of histogram

A histogram helps us understand:

- Whether an image is dark or bright.
- How pixel values are distributed.
- The contrast of an image.
- How an enhancement technique changes the image.

Comparing the original and enhanced histograms helps analyze the effect of image enhancement techniques.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Gamma Transformation

Gamma transformation is used to change the brightness of an image.

It is a nonlinear intensity transformation.

The gamma value controls how the brightness of the image is changed.

When the gamma value is less than 1, the darker regions of the image are mainly brightened. This can help reveal details that are difficult to see in dark areas.

Main effects

- **Adjusts image brightness.**
- **Brightens dark regions when gamma is less than 1.**
- **Helps reveal hidden details.**
- **Provides nonlinear brightness adjustment.**

Log Transformation

Log transformation is used to enhance details in the dark regions of an image.

It increases the visibility of pixels having low intensity values.

At the same time, it reduces the effect of very bright pixels.

Main effects

- **Enhances dark areas.**

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

- Makes low-intensity details more visible.
- Reduces the effect of very bright regions.
- Improves visibility of hidden information.

Therefore, log transformation is useful when important information is present in the darker parts of an image.

Image Negative

Image negative is a simple gray-level transformation that reverses the brightness of an image.

In an image negative:

- Dark pixels become bright.
- Bright pixels become dark.

For an 8-bit grayscale image, the transformation is:

$$s = 255 - r$$

For example, if a pixel has intensity 50, its negative intensity becomes:

$$255 - 50 = 205$$

This transformation can make certain details easier to observe.

Comparison of Techniques

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Technique	Purpose	Main Effect
Contrast Stretching	Improve low contrast	Expands intensity range
Histogram Equalization	Improve contrast	Redistributes intensity values
Gamma Transformation	Adjust brightness	Nonlinear brightness adjustment
Log Transformation	Enhance dark regions	Improves visibility of low-intensity pixels
Image Negative	Reverse brightness	Dark becomes bright and bright becomes dark

Importance of Gray-Level Transformations

Gray-level transformations are important because they help improve the appearance and quality of grayscale images.

They can be used to:

- Improve image contrast.
- Adjust image brightness.
- Highlight important details.
- Make dark regions more visible.
- Analyze pixel intensity distribution.
- Make images easier to interpret.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

The experiment demonstrates these five techniques and compares the original and enhanced images along with their histograms.

Code

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

# =====

# IMAGE PATHS

# =====

image_path_1 = '/content/images.jpg'
image_path_2 = '/content/images (1).jpg'
image_path_3 = '/content/images (2).jpg'
image_path_4 = '/content/images (3).jpg'
image_path_5 = '/content/images (4).jpg'

# =====

# READ IMAGES IN GRAYSCALE

# =====

src_image_1 = cv2.imread(image_path_1, cv2.IMREAD_GRAYSCALE)
src_image_2 = cv2.imread(image_path_2, cv2.IMREAD_GRAYSCALE)
src_image_3 = cv2.imread(image_path_3, cv2.IMREAD_GRAYSCALE)
src_image_4 = cv2.imread(image_path_4, cv2.IMREAD_GRAYSCALE)
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

```
src_image_5 = cv2.imread(image_path_5, cv2.IMREAD_GRAYSCALE)
```

```
# =====
```

```
# CHECK IF IMAGES ARE LOADED
```

```
# =====
```

```
if src_image_1 is None:
```

```
    print("Image 1 not found!")
```

```
    exit()
```

```
if src_image_2 is None:
```

```
    print("Image 2 not found!")
```

```
    exit()
```

```
if src_image_3 is None:
```

```
    print("Image 3 not found!")
```

```
    exit()
```

```
if src_image_4 is None:
```

```
    print("Image 4 not found!")
```

```
    exit()
```

```
if src_image_5 is None:
```

```
    print("Image 5 not found!")
```

```
    exit()
```

```
# =====
```

```
# 1. CONTRAST STRETCHING - IMAGE 1
```


 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

=====

```
min_val = np.min(src_image_1)
```

```
max_val = np.max(src_image_1)
```

```
if max_val == min_val:
```

```
    contrast_image = src_image_1
```

```
else:
```

```
    contrast_image = np.uint8(
```

```
        (src_image_1.astype(np.float32) - min_val)
```

```
        * 255.0 / (max_val - min_val)
```

```
    )
```

=====

2. HISTOGRAM EQUALIZATION - IMAGE 2

=====

```
histogram_image = cv2.equalizeHist(src_image_2)
```

=====

3. GAMMA TRANSFORMATION - IMAGE 3

=====

```
gamma = 4
```

```
normalized_image = src_image_3 / 255.0
```

```
gamma_image = np.power(
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

```

        normalized_image,
        gamma
    )

    gamma_image = np.uint8(
        gamma_image * 255
    )

# =====

# 4. LOG TRANSFORMATION - IMAGE 4

# =====

image_float = src_image_4.astype(np.float32)

c = 255.0 / np.log(
    15 + np.max(image_float)
)

log_image = np.uint8(
    c * np.log1p(image_float)
)

# =====

# 5. IMAGE NEGATIVE - IMAGE 5

# =====

negative_image = 255 - src_image_5

```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

```
# =====
# DISPLAY ORIGINAL AND ENHANCED IMAGES
# =====
```

```
plt.figure(figsize=(12, 15))
```

```
# Image 1 - Contrast Stretching
```

```
plt.subplot(5, 2, 1)
```

```
plt.imshow(src_image_1, cmap='gray')
```

```
plt.title("Image 1 - Original")
```

```
plt.axis('off')
```

```
plt.subplot(5, 2, 2)
```

```
plt.imshow(contrast_image, cmap='gray')
```

```
plt.title("Image 1 - Contrast Stretching")
```

```
plt.axis('off')
```

```
# Image 2 - Histogram Equalization
```

```
plt.subplot(5, 2, 3)
```

```
plt.imshow(src_image_2, cmap='gray')
```

```
plt.title("Image 2 - Original")
```

```
plt.axis('off')
```

```
plt.subplot(5, 2, 4)
```

```
plt.imshow(histogram_image, cmap='gray')
```

```
plt.title("Image 2 - Histogram Equalization")
```

```
plt.axis('off')
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Image 3 - Gamma Transformation

```
plt.subplot(5, 2, 5)
```

```
plt.imshow(src_image_3, cmap='gray')
```

```
plt.title("Image 3 - Original")
```

```
plt.axis('off')
```

```
plt.subplot(5, 2, 6)
```

```
plt.imshow(gamma_image, cmap='gray')
```

```
plt.title("Image 3 - Gamma ( $\gamma=0.4$ )")
```

```
plt.axis('off')
```

Image 4 - Log Transformation

```
plt.subplot(5, 2, 7)
```

```
plt.imshow(src_image_4, cmap='gray')
```

```
plt.title("Image 4 - Original")
```

```
plt.axis('off')
```

```
plt.subplot(5, 2, 8)
```

```
plt.imshow(log_image, cmap='gray')
```

```
plt.title("Image 4 - Log Transformation")
```

```
plt.axis('off')
```

Image 5 - Negative

```
plt.subplot(5, 2, 9)
```

```
plt.imshow(src_image_5, cmap='gray')
```

```
plt.title("Image 5 - Original")
```

```
plt.axis('off')
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

```
plt.subplot(5, 2, 10)

plt.imshow(negative_image, cmap='gray')

plt.title("Image 5 - Image Negative")

plt.axis('off')


plt.show()
```

Output :

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
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Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Image 1 - Original



Image 1 - Contrast Stretching



Image 2 - Original



Image 2 - Histogram Equalization



Image 3 - Original



Image 3 - Gamma ($\gamma=0.4$)



Image 4 - Original



Image 4 - Log Transformation



Image 5 - Original



Image 5 - Image Negative



 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Observations

1. Contrast Stretching – Image 1

- The original image had a limited range of intensity values, so the contrast was comparatively low.
- Contrast stretching expanded the intensity range toward **0–255**.
- Dark regions became clearer and bright regions became more distinct.
- The overall image appeared **sharper and more visible**.
- Therefore, contrast stretching improved the overall contrast and clarity of the image.

2. Histogram Equalization – Image 2

- The original image had an uneven distribution of intensity values.
- Histogram equalization redistributed the pixel intensities.
- This increased the overall contrast of the image.
- Details that were less visible in the original image became easier to observe.
- Therefore, histogram equalization improved the **visibility and contrast** of the image.

3. Gamma Transformation – Image 3

- Since the gamma value is greater than 1, the transformation makes the image **darker**, especially affecting the brighter intensity levels.
- This can improve visibility when excessive brightness is hiding details.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

- However, if the original image is already dark, this operation may reduce visibility instead of improving it.
- Therefore, the effectiveness of gamma transformation depends on the brightness of the original image.

4. Log Transformation – Image 4

- Log transformation changes the intensity values nonlinearly.
- It gives greater emphasis to lower-intensity regions.
- Darker details become more visible compared with the original image.
- Very bright regions have a relatively reduced effect.
- Therefore, the transformation is useful for revealing information present in darker areas.

5. Image Negative – Image 5

- The negative transformation reverses the intensity values.
- Dark regions become bright and bright regions become dark.
- This change can make certain features easier to distinguish.
- It provides an alternative view of the image and can highlight details that are less noticeable in the original.
- Therefore, image negative improves visibility for some types of images.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Image Processing	
Open Ended Problem 7	Date: 14/08/2026	Enrolment No:92400133050

Limitations

1. Results depend on the **input image**.
2. Different images may need **different enhancement methods**.
3. Fixed values like **gamma** may not work well for every image.
4. Too much enhancement can **reduce image quality**.
5. Histogram equalization may sometimes **over-enhance** the image.
6. The experiment uses only **grayscale images**.
7. Image negative is not useful for **every type of image**.
8. Results are mainly checked **visually**.

Conclusion

The experiment successfully demonstrated different gray-level transformation techniques for image enhancement. Contrast stretching, histogram equalization, gamma transformation, log transformation, and image negative were used to improve the visibility and contrast of images. The comparison of original and enhanced image helped analyze the changes in pixel intensity distribution. Overall, gray-level transformations are simple and effective techniques for improving image quality.